A UNIVERSAL PROPERTY OF THE FLATTENING IDEAL FOR FINITE TYPE ALGEBRAS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Base change behaviour of the flat-at-primes-over condition. Lemma \ref{lemma-base-change-flat-at-primes-over}, cited at proof line 4524. Gives one implication.
- The flat-at-primes condition can be descended along a directed colimit of rings. Lemma \ref{lemma-limit-preserving-flat-at-primes-over}, cited at proof lines 4533-4535. Moves the condition to a single essentially finite type subalgebra.
- The finite type version of the characterisation of the flattening ideal. Lemma \ref{lemma-flattening-complete-local-noetherian-property-by-finite-type}, cited at proof lines 4536-4538. Concludes that the ideal dies in the subalgebra.

Source reference. Original lemma 0528: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6561–6561. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0526: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6559–6559. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 052W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6584–6584. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 052X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6585–6585. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05LM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7241–7241. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0527: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6560–6560. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *With notation $R \to S$, M, and I and assumptions as in Lemma 0526. In addition assume that $R \to S$ is of finite type. Then for any local homomorphism of local rings $\varphi : (R, \mathfrak{m}) \to (R', \mathfrak{m}')$ the following are equivalent*

(1) *condition (052W) holds for $(R' \to S \otimes_R R', \mathfrak{m}', M \otimes_R R')$, and*
(2) *$\varphi(I) = 0$.*

Proof. The implication (2) $\Rightarrow$ (1) follows from Lemma 052X. Let $\varphi : R \to R'$ be as in the lemma satisfying (1). As R is Noetherian we see that $R \to S$ is of finite presentation and M is an S-module of finite presentation. Write $R' = \operatorname{colim}_\lambda R_\lambda$ as a directed colimit of local R-subalgebras $R_\lambda \subset R'$, with maximal ideals $\mathfrak{m}_\lambda = R_\lambda \cap \mathfrak{m}'$ such that each R_λ is essentially of finite type over R. By Lemma 05LM we see that condition (052W) holds for $(R_\lambda \to S \otimes_R R_\lambda, \mathfrak{m}_\lambda, M \otimes_R R_\lambda)$ for some λ. Hence Lemma 0527 applies to the ring map $R \to R_\lambda$ and we see that I maps to zero in R_λ, a fortiori it maps to zero in R'. □